



LAB 5

CONDITIONS AND LOOPS

LAB 5

- Review LAB on the concepts of chapters 3 and 4 of the book
- If you want, you can start from the exercises left behind in LAB04
- **RECOMMENDED EXERCISES**
 - **5.1.3**
 - **5.2.1**
 - **5.2.4**

Exercise 5.1.3

- Read request from Lab text..

Exercise 5.1.3 (cont'd)

- We want ALL factors, possibly repeated
 - If $N=8$ it must print ' 2 ' three times
- Basic algorithm:
 - value* = value read in input
 - Divide value* by all divisors starting from 2 until value =1
- *NOTE: while a divisor divides value , keep dividing, otherwise move on to the next divisor*
- A double cycle is needed !!!
 - Outer loop = the one on all divisors (until value >1)
 - Inner loop = until a certain divisor divides the value

Exercise 5.1.3 (cont'd)

Execution example: **value = 12**

Divisor = 2

value/divisor => SI **value =6** -> factor =2

value/divisor => SI **value =3** -> factor =2

value/divisor => NO **value=3** -> move to next divisor

Divisor = 3

value/divisor => SI **value =1** -> factor =3

Factors = 2,2,3

Exercise 5.2.1

- Read request from Lab text..

Exercise 5.2.1 (cont'd)

- We just need to apply the indicated formula 100 times.
 - Initially r_{old} is entered by the user
 - For each iteration we print r_{new}
- The only precaution: At each iteration what was r_{new} will become r_{old} in the next one!

Exercise 5.2.4

- Read request from Lab text..

Exercise 5.2.4 (cont'd)

- It is a matter of evaluating the formula

$$P_s = R_s \left(\frac{nV_s}{n^2 R_0 + R_s} \right)^2$$

with different values of n in steps of 0.01, and storing the value that gives the maximum P_s

- According to the figure, $V_s=40$, $R_0=20$, $R_s=8$